

SOUJANYA CHAVAN

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EDUCATION

University of Southern California

Master of Science in Analytics (GPA: 3.54/4.00)

Los Angeles, CA

Aug. 2024 – May 2026

○ **Coursework:** Applied Generative Artificial Intelligence, Predictive Analytics, Data Mining

A. P. Shah Institute of Technology, Mumbai University

Bachelor of Engineering in Computer Science (GPA: 3.67/4.00)

Mumbai, India

May 2020 – May 2024

○ **Coursework:** Artificial Intelligence, Machine Learning, Deep Learning, Database Management Systems

TECHNICAL SKILLS

AI-Assisted Development: Claude Code, Cursor, GitHub Copilot for daily agentic coding, context structuring, prompt design, **agent output verification**; LangGraph, LangChain, MCP-style tool schemas, RAG (ChromaDB, FAISS)

Languages: Go, TypeScript, JavaScript, Python, SQL; familiar with Java and Kotlin (coursework)

Distributed Systems & Async Pipelines: RabbitMQ, Celery, background jobs, DLQ retries, durable queues, manual acknowledgments, structured tracing, circuit breakers, timeouts, health checks, load testing (k6)

Databases & Concurrency: PostgreSQL (row-level locking, SELECT FOR UPDATE, optimistic CAS, transaction isolation, advisory locks, indexing), SQLite, MySQL, Redis (in-memory cache, distributed locks); ChromaDB, FAISS

Cloud, Containers & CI/CD: Docker, Docker Compose, **Kubernetes** (deployment/service manifests, rolling updates), **GitHub Actions CI/CD**; AWS (S3, Athena, SageMaker, Bedrock), GCP (Vertex AI, BigQuery, GCS), Azure; Git, Linux

Full-Stack Web & Observability: React, Next.js, Vite, Tailwind CSS; Node.js/Express, FastAPI, Pydantic, SQLAlchemy, Prisma, Socket.IO, WebSockets, REST APIs; dashboards, **structured JSON logging**, API logging, model monitoring

Testing & Evaluation: pytest, Playwright, Go test, unit + integration + concurrency contention tests, LLM evaluation harnesses, PyTorch, Hugging Face (A100 GPU training)

EXPERIENCE

Borealis Group Analytics

AI Engineer Intern

Blythewood, SC

July 2025 – December 2025

○ Shipped an internal AI Research Copilot behind a **FastAPI** service with run IDs, structured JSON logs, per-stage tracing, tool timeouts, and fallback responses for traceable failures.

○ Hardened tool routing with typed schemas, SQL guardrails, and **circuit breakers**, holding runtime errors under 3% on a 1,000-request load test.

○ Deployed a RAGAS CI gate across 250 QA pairs, lifting faithfulness 0.71 to 0.86 and context precision 0.64 to 0.81.

USC Information Sciences

Machine Learning Research Assistant

Los Angeles, CA

October 2024 – May 2025

○ Built a **RabbitMQ** async inference pipeline (durable queues, manual acks, retry queues, DLQ) processing 50K+ records/month into PostgreSQL for a FastAPI model service.

○ Scaled throughput 1,200 to 3,600 records/hour by parallelizing preprocessing and inference workers while keeping record-level failure recovery.

○ Fine-tuned RoBERTa with PEFT/LoRA adapters on **A100 GPUs**, lifting held-out accuracy 78% - 86% on 5K validation examples.

Motleyscape

Software Engineer

India

August 2023 – June 2024

○ Built an insights portal (**React/TypeScript** + Node.js/Express + PostgreSQL/Prisma + Redis) with dashboards and Socket.IO status updates for long-running jobs.

○ Cut p95 dashboard latency 1.8s to 420ms via Postgres indexes, Prisma query shaping, and Redis caching across 2K requests.

○ Owned Docker, **Kubernetes**, and GitHub Actions CI/CD with rolling-update manifests, enabling an A/B test that lifted task completion 63% to 80%.

PROJECTS

HotSeat: Single-Seat Concurrency Benchmark

August 2026 – September 2026

○ Built HotSeat (**Go**, PostgreSQL, Redis, k6, Docker), a concurrency benchmark implementing a single-seat booking flow four ways to see which survive 500 simultaneous requests, deliberately omitting the DB unique constraint so locking strategy was the only thing preventing a double booking.

○ Caught the harness producing a false pass on cold connection pool; rebuilt assertions across 5 rounds against an append-only ledger.

○ Published the unflattering Redis result honestly: correctness held, but 464/500 callers got HTTP 503 and p99 worsened 3.5x.

Handspan: Computer-Use Agent System for Legacy Enterprise UIs

July 2026 – September 2026

○ Built Handspan (**TypeScript**, Playwright, Claude Opus 5, **Claude Code**): an LLM discovers UI flows, successful runs compile into typed Zod-validated capability artifacts that replay deterministically with zero model calls.

○ Wrote 98 tests plus a **verify-outcomes eval harness** that caught a silent bug where model-emitted regex flags had killed 0/8 detectors.

○ Built a policy engine (allowlist, risk classifier, PII redaction, session-lease HITL handoff) blocking state-committing actions.

Multi-Agent GenAI Analytics System

October 2025 – December 2025

○ Architected a **LangGraph** supervisor routing between a RAG analyst and a slide generator with LLM plus rule-based fallback, backed by a FastAPI service with async Celery jobs on Redis/RabbitMQ.

○ Cut per-query LLM input 99.1% (50,890 to 476 tokens) via a ChromaDB RAG pipeline retrieving only top-5 rows.

○ Automated a 3,200-call factorial eval grid across a 4-provider fallback (OpenAI/Groq/Ollama/mock) at zero API cost.

ACHIEVEMENTS

Honors in AI/ML: University-conferred Honors in Artificial Intelligence & Machine Learning on the undergraduate degree.

Team Leadership: Led the team through project delivery during the AI Engineer internship at Borealis Group Analytics.

Hiring Contribution: Interviewed an intern candidate for Borealis, contributing to the evaluation and final hiring decision.